

Week 3 - Home and City

3

The Internet of Things (IoT)

R&D

The vision of a future taken by networked
algorithmic advancements

"Smart cities" grow into

→ reality is often expensive / unreliable systems

- fail to integrate
- frustration
- poor quality data
- arguments on surveillance
cost cutting

Gov → control
↓
" regulations

Can Gov / organisations be
trusted?

No privacy
No freedom

Mistake?

Dr James Stewart + Dr Zachary Urguhart

⇒ Data → how do we ensure it is used in a lawful way?

Who uses it / handles

" logs "

Who stores it

" collects "

The impact this DATA has on society

"Contextual integrity" - info moved out of context
↓
privacy harmed

Data collection is lawful - in cities - how do people consent?
are aware?

GDPR - new legislation on data protection
- new obligation like doing data protection by design and default
developers → dealing with regulators

→ Risk to get products in the market means they might not be
secure (re data)
→ leak data
→ we don't have the skills to make sure they're secure
→ quality of infrastructure?

professional practice
regulation
legislation

← how do we develop

↓
to make sure whoever develops something think of privacy security
consequence

Privacy by design and default → get designers think of developers

HCI (human computer interaction) design ethnographies
participatory design
value sensitive design

→ ^{decide as} ~~believe~~ we are ~~on~~ a society re how we are surveyed

→ function creep: where you set up X to do one job and as they change the backend software and what you look for that data

↓
you can do new things with it

use the data for something $\neq$ (that ^{even} wasn't consented for)

Products
+
Systems } for private homes

ex. voice interfaces ← Amazon Echo } use IoT devices
Google Home } AI system
↓
provides intelligent control
automated
•
over our lives

Our data ← stored all over
sold → for other services
|

if someone in the family wants to use a device
reusing the data

|
might not be consensual w/ the family
" " realise what they agree to

|
corporations → force others to adopt it
gors

CONCERNS

Social and cultural

- expectations to be safe at home (hacked Amazon device)
- tech can be problematic mainly for vulnerable groups
- users become ↓ aware of tech: they're embedded
- inertia and convenience — they "want" to erase their data but they don't

|
privacy paradox

- Data breaches ← software hijacking

Governance

legal frameworks → EU General Data Protection Regulation

↓
responsibilities } around personal data
rights

DATA:

- Who has access
for what it's used
- where is stored
for how long

those who (have to)

Collect y data → put in place technical safeguards to protect
govern organisational personal data

+

communicate to users

problem → users don't read T&C
but cannot renegotiate them

↓
in some senses you can't see what you consented for
~~that~~ withdraw consent

Prof Helen Nissenbaum's theory of privacy

↓
it involves protecting the contextual integrity of
informational flows

Prof Danah Boyd → children don't care companies knowing their
social media activities / but they care ^{readers knowing} about

The Smart City

Ex Sidewalks Lab (Toronto)

- Concerns about digital firms buying public space → running them w/ "market logic"
- worries about residents' privacy rights
- uncertainty — how much say residents have over the design of the space over what technologies are embedded in the environment

Smart Cities

→ Constraints — cost → public/private partnerships can fill this gap

complexity of data ← collections
sharing
action
generated
interpreted

• Live Facial Recognition

What safeguards are necessary for the public?

→ Data protection impact assessment

↓ can

may risks

necessary safeguards →

practical challenges in implementation

Individuals rights to privacy vs. national security
Groups + detect/grow/track/investigate
crime

→ Recognition of facial emotions.

Reinforcing inequalities

concerns of → Unequal deployment and use
systematic replacement of human decision making for
algorithms

example → for improve traffic in 1 area
to restrict freedom in another

! function creep

- Security and misuse
wrong hands

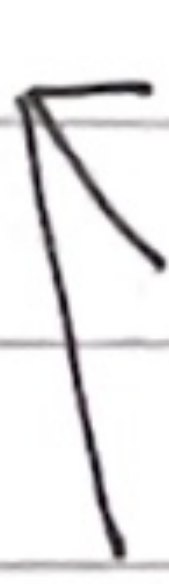
SOLUTIONS

→ promote public engagement

→ Opening access → challenges:

- Creating + Maintaining datasets → costly
- anonymise data → difficult
- ensure harmonised standards of data management → difficult
- selective release for political gain
- impact neighbourhoods / house prices
- promote & biases

Value Sensitive Design - term popularized by Prof Batya Friedman

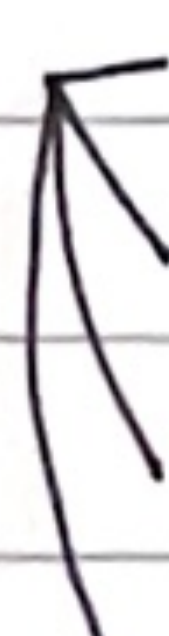
who builds a tech can be  technical
social
political matter

Prof Lawrence Lessig
"code is law"

↓

architecture of tech can have the functional equivalence of law when trying to regulate user behaviour

Values w/ ethical import

 privacy
democracy
freedom of speech
equality

should be
suggested by
tech

• fog and edge computing